

Chapter 7 Programming Software

Programming software allows programmers to develop new **programs**. The most basic programming software is a **source code editor**. These programs are usually basic **text editors**. Programmers use them to enter lines of code into a computer. Lines of code are written in a **programming language**. Programmers use this to write series of commands. When executed *together*, these commands form a complete program.

Other **programming software** are more complex. An **IDE** (integrated development environment) provides tools for writing programs. These usually include a text editor and a **debugger**. Other tools may include a **linker**, a **compiler**, or an **interpreter**.

Get ready!

- ① Before you read the passage, talk about these questions.

- 1 How do programmers create new programs?
- 2 What kinds of software helps programmers write new programs?

Reading

- ② Read the textbook chapter. Then, mark the following statements as true (T) or false (F).

- 1 ☐ Programmers use text editors to interpret commands.
- 2 ☐ Text editors are a type of source code editor.
- 3 ☐ An IDE is the most basic type of programming software.

1 ☐ source code editor

2 ☐ program

3 ☐ programming language

4 ☐ programming software

5 ☐ interpreter

6 ☐ compiler

A any software that supports the development of new applications

B a series of operations that control the functions of a computer

C codes used to write commands to a computer

D an application that decodes instructions written in other languages

E something that reads and executes other programs

F software used to enter lines of coded text

- ➊ Read the sentence pairs. Choose which word or phrase best fits each blank.

1 **text editor / IDE**

- A A(n) _____ usually provides programmers with various tools for writing programs.
 B A(n) _____ is a basic program for entering commands and code into a computer.

2 **debugger / linker**

- A A _____ provides links to additional information needed for programs to run.
 B A _____ finds and corrects errors in code.

- ➋ Listen and read the textbook chapter again. What tools does an IDE provide?

Listening

- ➌ Listen to a conversation between two students. Choose the correct answers.

- What is the conversation mostly about?
 - how to interpret instructions in programming code
 - the difference between two types of computer programs
 - techniques for entering code into a text editor
 - the importance of learning different programming languages
- According to the woman, what is sometimes needed to run programs correctly?
 - using a linker and a compiler together
 - allowing programs to connect to the internet
 - writing new instructions in a programming language
 - translating lines of code into a different programming language

- ➍ Listen again and complete the conversation.

Student 1: Can you help me with something?

Student 2: Sure. What is it?

Student 1: I'm confused 1 _____ between linkers and compilers.

Student 2: Okay. Well, 2 _____ a program to information it needs from other programs.

Student 1: Wait, I thought an 3 _____ did that.

Student 2: No. An interpreter just 4 _____ so the program can run.

Student 1: Okay. So then a compiler is 5 _____ program instructions.

Student 2: Exactly. It translates information from different 6 _____

Student 1: I get it now. Thanks.

Speaking

- ➎ With a partner, act out the roles below based on Task 7. Then, switch roles.

USE LANGUAGE SUCH AS:

Can you help me with ...
 I'm having some trouble ...
 I was under the impression ...

Student A: You are a student. Talk to Student B about:

- terms you are confused about
- what you thought the terms meant
- what the terms actually mean

Student B: You are a student. Talk to Student A about terms that he or she is confused about.

Writing

- ➏ Use the conversation from Task 8 to complete the workbook exercise.

Explain the difference between each set of terms:

- 1 Debugger / _____:

A debugger is used for finding and correcting errors in code.

A(n) _____ is used for _____.

- 2 _____ / text editor:

A(n) _____ is used for _____.

A text editor is used to enter lines of code and create programs.